

## PLANNING (first stage, annualized CAPEX)

fleet composition & battery class · FFH specification  
charger siting & power class · site activation  
grid-connection tier



capacities, fleet,  
FFH availability

expected OPEX,  
feasibility



## OPERATIONS (recourse per scenario $\omega$ )

bus-class  $\leftrightarrow$  block assignment  
charging: TOU · plug ratings · site & grid caps  
heat dispatch: electric vs FFH · preconditioning  
arrival-SoC reserve · overnight cyclic closure  
cold-derated capacity & charging acceptance



### Identified loads

thermal model [1]  
traction fit  $e(T, v)$

### Scenarios $\omega$

ECCC bands  
 $\times$  day counts  $h_\omega$

### Objectives

min cost · CO<sub>2</sub>:  
price  $\lambda$  or cap  $\varepsilon$